

Cascading Spy Sheets: Exploiting the Complexity of Modern CSS for Email and Browser Fingerprinting

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In an attempt to combat user tracking, both privacy-aware browsers (e.g., Tor) and email applications usually disable JavaScript. This effectively closes a major angle for user fingerprinting. However, recent findings hint at the potential for privacy leakage through selected Cascading Style Sheets (CSS) features. Nevertheless, the full fingerprinting potential of CSS remains unknown, and it is unclear if attacks apply to more restrictive settings such as email.

In this paper, we systematically investigate the modern dynamic features of CSS and their applicability for script-less fingerprinting, bypassing many state-of-the-art mitigations. We present three innovative techniques based on fuzzing and templating that exploit nuances in CSS container queries, arithmetic functions, and complex selectors. This allows us to infer detailed application, OS, and hardware configurations at high accuracy. For browsers, we can distinguish 97.95% of 1176 tested browser-OS combinations. Our methods also apply to email applications - as shown for 8 out of 21 tested web, desktop or mobile email applications. This demonstrates that fingerprinting is possible in the highly restrictive setting of HTML emails and expands the scope of tracking beyond traditional web environments.

In response to these and potential future CSS-based tracking capabilities, we propose two defense mechanisms that eliminate the root causes of privacy leakage. For browsers, we propose to preload conditional resources, which eliminates feature-dependent leakage. For the email setting, we design an email proxy service that retains privacy and email integrity while largely preserving feature compatibility. Our work provides new insights and solutions to the ongoing privacy debate, highlighting the importance of robust defenses against emerging tracking methods.

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